

# Arthur Lau

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## Profile Summary

A **Results-driven** Software Engineer with 7+ years of experience in **full-stack** development specializing in **React, Next.js, Python, C#(.Net) Flutter, Cloud(Azure, AWS)** and **vibe coding**. **Adaptable** in delivering **scalable, secure, and robust** solutions across diverse domains such as vending automation, ERP, IoT, finance and gaming with proficiency in designing **system architecture**, implementing **performance optimization** and driving **project accountability**. Strong in **cross-functional communication** with PMs, analysts and engineers to yield business-aligned outcomes. Consistently pursue **technical craftsmanship** through team mentorship, and maintainable code practices.

## Experience

### Senior Software Engineer, IVM Tech Limited

Aug 2023 – Present

A leading vending machine company in the domestic market, serving 2k+ sellers and 500k+ customers.

- Spearheaded the first weighted vending machine system using **Next.js** with **Prisma, Flutter**, and **MSSQL, SocketIO**, reaching 250K+ users.
- Integrated **secure** payment gateway with **POS** machine by **Java** and **Kotlin**, also **Wechat** and **Alipay** on **C#** for more than 1000+ clients.
- Built real-time admin portal for 10k+ store managing \$1M+ transaction by **NestJS** and **MSSQL, EJS, HTML, CSS** and **jQuery**.
- Automated **CI/CD** on **AWS** and **Azure** with comprehensive testing using **Vitest, MStest** and **Playwright**, minimizing deployment time by 60%.
- Designed specifications across **design** and **operation** teams with 10+ smart fridge partners ensuring consistent interoperability.
- Led ERP modernization by migrating to **.NET** microservices with **MSSQL, MQTT, Redis, React** and **Tailwind**, with **Azure DevOps**.

### Software Engineer, Makeitfun Studio

Jul 2022 – Aug 2023

A digital creative studio providing fun experiences via various gamification forms with an average of 100k+ active users.

- Constructed **Responsive SEO**-optimized **React + Next.js** event websites serving 500k+ users, boosting reach by 45%.
- Designed scalable db and logic with **RESTful** and **GraphQL** for 10+ projects via **ExpressJS, MongoDB, Postgres, docker** and **Kubernetes**.
- Delivered a 3D real-time voting system from **Mozilla Hubs'** open-source metaverse platform, using **React, Three.js, Elixir, WebRTC**, and **Express.js**
- Programmed an **AR Flutter** app for a Disney-affiliated Studio using **MediaPipe** and **Vuforia** in **Unity** and **C#** with 70k+ served customers.

### Software Engineer, Flexsystem Limited

Jun 2021 – Jul 2022

An international provider of enterprise management SaaS solutions for 2000+ clients in 30+ countries.

- Tailor-made an **SaaS ERP** system for International client with over 2M+ users using **React, Angular, C#** and **ISAM**.
- Created **RPA** by **UIPath** for **stress** and **unit** tests with **Jest** and **Selenium**, cutting repetitive task execution time by 50%.

### Software Engineer, Excohk Limited

Oct 2020 – Jun 2021

A start-up cultural media collective that bridges art, storytelling, and creative branding for 50+ institutions and communities.

- Developed an **OAuth2 booking** system from **React Native, Expo, Java EE**, enabling personalized event experiences on 50+ large-scale cultural events.
- Implemented a **temporal storage solution** via **DynamoDB** and **Firebase**, improving responsiveness and visitor interaction reliability.

### Full Stack Engineer, East Technology Limited

Jul 2020 – Sep 2020

An award winning media provider delivering content display, e-commerce solution for 180+ brands

- Implemented UX on SPA website by **PHP, jQuery**, and **JavaScript** across 20+ business and educational domains.
- Developed a lightweight **CMS** using **PHP, Apache**, and **phpMyAdmin**, enabling customized page updates for non-technical users.

### Backend Developer, Yunfeng Financial Group Limited

Jun 2019 – Jul 2019

A leading financial company across Asia market cooperating with 1000+ of global partners

- Programmed and Unit-tested **Flask** Server API of Youyu Smart Trade App using **Python** towards **MySQL**, ensuring production level reliability.

## Projects

[Prompt Pounder](#) - Coded a game by **JavaScript, HTML** and **CSS** to retain focus during AI prompt delays, reducing distractions.

[Arkham Analysis](#) - Building a data-analytic game assistant for Arkham Horror LCG with 1000+ cards by **FastAPI, PostgreSQL**, and **Angular**.

[Reals](#) - Created an AI-powered **Flutter** app for custom dating itineraries with **ChatGPT** and **FastAPI**, providing click-to-plan experience.

[Mozilla Hubs](#) - Integrated open source 3D chess with Avatar Maker and observer mode using **React, Three.js, Elixir, WebRTC, Express.js**

## Skills

Programming Languages: **C#, JavaScript, TypeScript, Java, Python, Dart, Kotlin, PHP, Erlang, Elixir**

Programming Practise: Design Pattern, Architecture(MVC, microservice, even-driven etc), **SOLID**, version control, optimization, Agile

Back-end Development: **.NET Core, ASP.NET, Node.js, Express.js, RESTful APIs, Django, Flask, FastAPI, Spring Boot, GraphQL, Go**

Front-end Development: **React, Next.js, Flutter, NestJS, Angular, Tailwind CSS, Material-UI (MUI), Bootstrap, Vue, Solid,**

Database Management: **PostgreSQL, MSSQL, SQLite, MongoDB, Redis, MySQL, DynamoDB, Firebase**

DevOps & Cloud Platforms: **Git, GitHub, GitLab, VS, TortoiseSVN, Docker, Kubernetes, Jenkins, UIPath, Azure, AWS, Cloudflare, GCP.**

## Certificate

**AWS Certified Solution Architect (In Progress)**

## Education

**Chinese University of Hong Kong** – Bachelor's Degree in Computer Science

Jun 2021